

Checkpoint

14.2: Limits and Continuity

Problem 1. *What is $z = f(x, y)$? How many independent and dependent variables does it have?*

Problem 2. *Why is it harder to find determine whether a limit exists on a surface than on a curve?*

Problem 3. *Find $\lim_{(x,y) \rightarrow (1,1)} e^x y + 2xy - y^2$*

Problem 4. *Is the following statement valid or invalid?*

“The $\lim_{(x,y) \rightarrow (a,b)} f(x, y)$ exists and is equal to c since along the path $x = a$, $f(x, y)$ approaches c , and along the path $y = b$, $f(x, y)$ approaches c ”

Problem 5. *Determine if $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2+y^4}$ exists or does not exist.*

1 Answers

1. A surface with one dependent variable z and two independent variables x, y .
2. There are infinitely many paths along my independent variables to get to the desired point (x, y) whereas for a curve there are just two (left and right).
3. $e + 1$
4. Invalid, to show a limit does exist we need to prove along *all* paths, $f(x, y)$ approaches c .
5. Does not exist.